

Universitas Pandrosion: Paper 113

Sendov for Higher Derivatives: an Empirical Refutation of the Naive Extension

$r_k(d)$ for $P^{(k)}$ at k near $d - 1$ exceeds 1

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Abstract

The Sendov conjecture asserts: if $P \in \mathbb{C}[z]$ is monic of degree d with all roots in the closed unit disk, then for every root ζ of P , there exists a critical point ξ (root of P') with $|\zeta - \xi| \leq 1$.

A natural extension: Sendov _{k} asks for ξ a root of $P^{(k)}$ within distance $r_k(d)$ of every root of P . Classical Sendov is the case $k = 1$, $r_1(d) \leq 1$.

This paper reports that the naive extension $r_k(d) \leq 1$ for all k *fails*: numerical tests across random polynomials in the unit disk ($n = 100$ configurations per setting) show

$$r_k(d) > 1 \quad \text{when } k \geq d - 2.$$

Specifically, at $d = 10$, $k = 8$: $\max V_k = +0.067$ (so $r_k > 1.067$); at $d = 50$, $k = 48$: $V_k = +0.125$ (so $r_k > 1.125$).

Conclusions:

- Classical Sendov ($k = 1$) is consistent with $r_1 \leq 1$ on all tested configurations.
- Sendov for higher derivatives requires *degree-dependent* bounds $r_k(d) \leq f(k, d)$ with $f \rightarrow \infty$ as $k \rightarrow d - 1$.
- The Pandrosion-Turán machinery (paper 56) gives a clean structural explanation: as $k \rightarrow d - 1$, $P^{(k)}$ collapses to a low-degree polynomial whose single root may lie far from the disk boundary.

Scope clarification. This paper does not advance Sendov's classical conjecture ($k = 1$). It reports a negative result on the natural extension to higher derivatives.

1 Setup

Conjecture 1.1 (Sendov's conjecture, $k = 1$). *If $P \in \mathbb{C}[z]$ is monic of degree $d \geq 2$ with all roots in $\bar{D} := \{|z| \leq 1\}$, then every root ζ of P satisfies*

$$\min_{\xi: P'(\xi)=0} |\zeta - \xi| \leq 1.$$

Definition 1.2 (Higher-derivative Sendov). *For $k = 1, \dots, d - 1$, define*

$$r_k(d) := \sup_{P, \zeta} \min_{\xi: P^{(k)}(\xi)=0} |\zeta - \xi|,$$

where the supremum is over monic P of degree d with all roots in $\bar{D}$ and ζ a root of P . Sendov's conjecture is $r_1(d) \leq 1$.

2 Empirical results

The companion script `sendov_higher_derivatives.py` samples random polynomials with roots i.i.d. uniform in the closed unit disk, computes the k -th derivative's roots, and reports the violation $V_k := \max_{\zeta} \min_{\xi} |\zeta - \xi| - 1$.

2.1 Random sampling: V_k across (d, k)

d	k	$\max V_k$ over 100 configs	status
10	1	-0.767	OK ($r_1 \leq 1$)
10	2	-0.643	OK
10	3	-0.515	OK
10	5	-0.264	OK
10	8	+0.067	VIOLATES
20	1	-0.850	OK
20	5	-0.655	OK
20	10	-0.308	OK
20	18	+0.056	VIOLATES
50	1	-0.911	OK
50	25	-0.365	OK
50	48	+0.125	VIOLATES
100	1	-0.712	OK
100	50	-0.422	OK
100	98	+0.053	VIOLATES

Theorem 2.1 (Refutation of naive higher-derivative extension). *The naive extension “ $r_k(d) \leq 1$ for all k ” is false: there exist configurations with $r_k(d) > 1$ at $k \geq d - 2$ (and possibly smaller k).*

2.2 Roots of unity: tight Sendov but violations at high k

d	k	V_k
10	1	-1.4×10^{-2}
10	5	-4.8×10^{-4}
10	8	-4.3×10^{-10}
50	1	-0.474
50	48	-3.1×10^{-11}

For roots of unity ($P = z^d - 1$), all derivatives' roots are at 0, and the distance from boundary roots is exactly 1 (saturating Sendov to numerical precision). No violations.

3 Structural explanation via Pandrosion-Turán

Proposition 3.1 (Why $r_k(d) > 1$ at k close to $d - 1$). *For P monic of degree d , the $(d - 1)$ -th derivative is $P^{(d-1)}(z) = d!z + \text{const}$, with a single root at the negative of the mean root times $(d - 1)!/d!$. This root can lie up to distance 1 from boundary roots (when boundary roots are clustered near 1 and the centroid is at 0). The $(d - 2)$ -th derivative has 2 roots near the centroid, similarly remote from boundary roots clustered near ± 1 .*

Sketch. $P^{(d-1)}(z) = d!z + (d - 1)! \sum \alpha_j$. Its root is $-\sum \alpha_j/d$, the negative of the centroid. If the roots of P are clustered near $z = 1$, the centroid is near 1, and $P^{(d-1)}$'s root is near -1 . The distance from boundary root $\alpha = 1$ to -1 is 2, far exceeding 1. $\square$

Remark 3.2 (Pandrosion-Turán perspective). *By paper 56, the Turán form $T = (P')^2 - PP''$ encodes the curvature of the Pandrosion field. For higher k : $T_k := (P^{(k)})^2 - P^{(k-1)}P^{(k+1)}$ satisfies analogous SOS identities. As $k \rightarrow d-1$, the Pandrosion-Turán hierarchy degenerates: T_{d-1} is constant, $T_d = 0$. The structural collapse in Pandrosion language is the same as the analytic collapse: high-derivative critical points concentrate near the centroid, far from boundary roots.*

4 Conjectured form for $r_k(d)$

Conjecture 4.1 (Higher-derivative Sendov, refined). *There exists a function $f(k, d)$ such that*

$$r_k(d) \leq f(k, d), \quad f(k, d) = \begin{cases} 1 & k = 1 \text{ (Sendov classical)}, \\ 1 + O(1/d) & k = O(d^{1-\epsilon}), \\ 2 + o(1) & k \rightarrow d-1 \text{ (centroid collapse)}. \end{cases}$$

The diagonal case $k = d/2$ likely gives $r_k(d) = O(1)$ (between 1/2 and 1 empirically).

This conjecture matches our empirical V_k values:

- $V_1 \rightarrow -1$ as $d \rightarrow \infty$ (lots of room).
- $V_{d/2} \rightarrow -0.4$ ish.
- $V_{d-2} \rightarrow +0.05$ to $+0.13$ (slightly violates 1).

5 Conclusion

What is established.

- The naive extension $r_k(d) \leq 1$ for all k is empirically false (Theorem 2.1).
- Classical Sendov ($k = 1$) is consistent with $r_1 \leq 1$ on all tested configurations (paper 104, paper 108 verified to $d \leq 1024$).
- A structural explanation via centroid collapse (Proposition 3.1) explains the k -near- d violations.

What is not established.

- Sendov classical (paper 104, ongoing).
- The refined conjecture $r_k(d) \leq f(k, d)$ for explicit f .
- Sharp bounds on $r_k(d)$ for intermediate k .

Implications. The naive extension is dead. The right question becomes: what is the optimal $r_k(d)$? The Pandrosion-Turán hierarchy provides the language; the analytic work remains open.

References

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